

YVETTE D. HASTINGS

+1(801)520-0991 • yvettehastings6@gmail.com

GitHub: [@HastingsY](#) • LinkedIn: [yvette-hastings-2a47231b/](#)

ORCID: [0009-0000-2143-5634](#) • Website: hastingsy.github.io/Design_Portfolio/

PERSONAL STATEMENT

Interdisciplinary researcher specializing in human-centric software quality, with a background spanning analytical chemistry, geospatial science, and computer science. Experienced in statistical modeling, R-based analysis, and software quality evaluation, with a focus on bridging human-centered metrics and technical systems. Proven record of peer-reviewed publications and applied research across environmental science and cybersecurity domains.

EDUCATION

- MS** Computer Science (Coursework Track)
Montana State University, Bozeman, MT, August 2026
Graduate Research Assistant (non-thesis program)
- MS** Geography
University of Utah, Salt Lake City, UT, August 2022
Thesis Title: “Green Infrastructure Microbial Community Response to Simulated Pulse Precipitation Events in the Semi-Arid Western U.S.”
NSF Grant: “Can green infrastructure maximize ecosystem processes related to nitrogen?” (NSF DEB 2006308)
Advisor: Dr. Jennifer Follstad Shah
Graduate Certificate in Hydrology and Water Resources
- BS** Environmental and Sustainability Studies
University of Utah, Salt Lake City, UT, July 2020
Certificate in Geographic Information Systems (GIS)
- BA** General Science
North Central College, Naperville, IL, May 2007
Minored in History

TECHNICAL SKILLS

Programming Languages: R, Python, SQL, Java, JavaScript, TypeScript
Tools: ArcGIS, ESRI Suite, R Shiny, React
Methods: Statistical modeling, data visualization, geospatial analysis, machine learning

HONORS AND AWARDS

Montana State University Graduate School University Fellowship, Spring 2023 – Spring 2026

Fellowship award based on academic and scholarship achievements to assist in the cost of living during graduate school.

Weber State University Intermountain Sustainability Summit Student Poster Competition, Spring 2022

Student poster, graduate student category, winner for the 2022 Intermountain Sustainability Summit. Poster submission titled: “Green Infrastructure Microbial Community Response to Simulated Storm Events in Semi-Arid Environments”.

Global Change and Sustainability Center Student Travel Award, Spring 2022

Student travel funding to the Consortium of Aquatic Science Societies Joint Aquatic Science Meeting in Grand Rapids, MI, in May 2022.

Teaching Assistantship and Research Assistantship, 2021 & 2022 Academic Year

Student tuition, teaching, and research assistantship during Master of Science studies provided by the University of Utah Environmental and Sustainability Program and Department of Geography.

Undergraduate Summer Research Assistantship, Summer 2004

Assist the Department of Chemistry junior faculty conduct summer research.

RESEARCH EXPERIENCE

Thesis, University of Utah, Salt Lake City, UT, 2020 – 2022

Graduate Research Assistant, Dr. Jennifer Follstad Shah

- Provide project management expertise to monitor research progress.
- Assist in project database management.
- Assist in protocol development, laboratory setup, and chemical hygiene monitoring.
- Utilize laboratory equipment to complete protocol extractions and assays.
- Perform statistical analysis on research results.

North Central College, Naperville, IL, 2004

Undergraduate Research Assistant, Dr. Jeffery Jankowski

- Investigate non-corrosive fuel additives for methanol fuel.
- Establish lab protocol and safety parameters for flammable materials used during research applications.

RESEARCH PUBLICATIONS

Y. D. Hastings et al. (2026), “Barriers to Use: Perspectives on Environmental Research Software,” in *21st International Conference on Software Technologies*. In press.

Y. D. Hastings and A. M. Reinhold (2026), “Software quality-in-use: A systematic literature review,” in *2026 Intermountain Engineering, Technology and Computing (iETC)*, IEEE, 2026.

Y. D. Hastings and A. M. Reinhold (2026), “From standards to practice: A position on challenges in operationalizing software quality-in-use,” in *21st International Conference on Evaluation of Novel Approaches to Software Engineering – Volume 1*, ISBN 978-989-758-828-0, ISSN 2184-4895, pp. 859-868.

Y. D. Hastings et al. (2026), “Mining user forums to evaluate quality-in-use of environmental software,” in *SoutheastCon 2026*, IEEE, 2026, pp. 1–8.

M. H. Munro, R. J. Gore, C. J. Lynch, **Y. D. Hastings**, and A. M. Reinhold (2025), “Enhancing risk and crisis communication with computational methods: A systematic literature review,” *Risk Analysis*, vol. 45, no. 7, pp. 1683–1697, 2025.

Y. D. Hastings et al. (2023), “Green infrastructure microbial community response to simulated pulse precipitation events in the semi-arid western united states,” *Water*, vol. 16, no. 13, p. 1931, 2024.

Y. D. Hastings and A. M. Reinhold (2023), “Applying software quality in use standards to improve scientific software selection,” *WiPiEC Journal-Works in Progress in Embedded Computing Journal*, vol. 9, no. 1, pp. 4–4, 2023.

PRESENTATIONS

Hastings, Y.D., & Reinhold, A.M. (2026, May 8). *Software quality-in-use: A systematic literature review* [Oral Presentation]. 2026 Intermountain Engineering, Technology and Computing, Provo, UT.

Hastings, Y.D., & Reinhold, A.M. (2026, March 13). *Mining User Forums to Evaluate Quality-in-Use of Environmental Software* [Oral Presentation]. SoutheastCon 2026, Huntsville, AL.

Hastings, Y.D. (2025, May 21). *Hands-on demo: Digital Forensics* [Invited Presentation]. 2025 Cybersecurity Education and Research Summer Workshop, Washington State University, Pullman, WA.

Follstad Shah, F., **Hastings, Y.D.**, Smith, R., & Goel, R. (2024, June 5). *Microbial community diversifies while physiological capacity diminishes in newly constructed stormwater bioswales of semi-arid Utah, USA*. Society for Freshwater Science, Philadelphia, PA. Presented by Follstad Shah, F.

Hastings, Y.D. (2024, March 15). *Environmental Software Engineering* [Invited Presentation]. Spatial Utah Data Science, University of Utah, Salt Lake City, UT.

Izurietta, C., Woods, N., & Reinhold, A.M. (2023, September 8). *A Brief of Distributed Data Processing* [Oral Presentation]. DSD-SEAA 2023 MicroEuropean Conference, Works in Progress Session, Durres, Albania. Presented by **Hastings, Y.D.** on behalf of the authors.

Hastings, Y.D., & Reinhold, A.M. (2023, September 8). *Applying Software Quality in Use Standards to Improve Scientific Software Selection* [Oral Presentation]. DSD-SEAA 2023 MicroEuropean Conference, Works in Progress Session, Durres, Albania.

Hastings, Y.D., Mann, K.A., Smith, R.M., & Follstad Shah, J (2022, November 16). *Green Infrastructure Microbial Response to Storm* [Poster Presentation]. Salt Lake County Watershed Symposium. <https://watershedsymposium2022.sched.com/artist/u1105374>

Hastings, Y.D. (2022, June 13). *Green Infrastructure Microbial Community Response to Simulated Pulse Precipitation Events in the Semi-Arid Western U.S.* [Thesis Defense]. University of Utah.

Hastings, Y.D., Mann, K.A., Smith, R.M., & Follstad Shah, J. (2022, May). *Green Infrastructure Microbial Community Response to Simulated Storm Events in Semi-Arid Environments* [Poster Presentation]. Consortium of Aquatic Science Societies Joint Aquatic Science Meeting.

Hastings, Y.D., Follstad Shah, J., Mann, K.A., & Smith, R.M. (2022, February). *Green Infrastructure Microbial Community Response to Simulated Storm Events in Semi-Arid Environments* [Poster Presentation]. University of Utah Environment and Sustainability Research Symposium. <https://www.environment.utah.edu/students/student-project/yvette-hastings/>

PROFESSIONAL DEVELOPMENT TRAINING

Reactive Transport with Sophisticated Reaction Networks

Pacific Northwest National Laboratory, Richland, WA, 2023

Description: CUAHSI training on reactive transport with hands-on experience using PFLOTRAN to solve environmental problems.

Reactive Transport Modeling Summer Institute

Colorado School of Mines, Golden, CO, 2023

Description: Training on reactive transport modeling with hands-on experience using CRUNCHTOPE to solve environmental problems.

Research Education

University of Utah, Salt Lake City, UT, 2020 – 2021

Mentor Training: Introduction to Research Mentoring; Assessing Understanding and Fostering Independence; Establishing Expectations and Maintaining Effective Communication

Research Courses: Annual Teaching Symposium; Getting Published: Responsible Authorship and Peer Review; Introduction to Responsible Conduct of Research; Rigor, Transparency, and Reproducibility in Research

PFC for Professionals

Bureau of Land Management, National Resources Conservation Service, U.S. Fish and Wildlife Service, Salt Lake City, UT, Summer 2021

Description: Classroom and field training on Proper Functioning Condition (PFC) assessments for federally managed lentic and lotic systems.

Meadow Restoration

Bureau of Land Management, National Resources Conservation Service, U.S. Fish and Wildlife Service, Cedar City, UT, Summer 2021

Description: Classroom training for meadow condition assessments. Identify impaired meadows and learn techniques to improve meadow conditions.

PROFESSIONAL WORK EXPERIENCE

Graduate Research Assistant

Montana State University, Bozeman, MT, January 2023 – Present

- Design and develop web-based visualizer for PIQUE model output.
- Contribute to Software Bill of Materials (SBOM) working group focused on software supply chain security.
- Mentor 4-5 undergraduate students in cybersecurity.
- Perform empirical investigations into software adoption barriers.

Teaching and Research Assistant

University of Utah, Salt Lake City, UT, August 2020 – December 2022

- Assist 200+ undergraduate students in achieving academic success in course objectives and final project deliverables.
- Perform ecology and hydrology research to understand surface water impacts on urban stream systems.

Hydrology Technician – Student Intern, Volunteer

Bureau of Land Management, Volunteer, Salt Lake City, UT, May 2019 – April 2022

Bureau of Land Management, GIS Internship, Salt Lake City, UT, Summer 2020

Bureau of Land Management, University of Utah Hinckley Internship, Salt Lake City, UT, Summer 2019

- Collect samples and perform field assessments of federal rangeland and riparian areas with team members to compile data for rangeland management programs.
- Perform riparian restoration projects with team members to revegetate fire-devastated rangeland.
- Generate, compile, interpret, and transfer GIS and field assessment data into National Proper Functioning Condition Database to prioritize riparian restoration projects.
- Beta test new GIS and digital assessment software to streamline field assessments.

Chemist, Inorganic Department Rover/Assistant Supervisor

American West Analytical Labs, Salt Lake City, UT, July 2016 – July 2020

- Train 30+ new and current employees on EPA and laboratory methods to enhance performance and efficiency.
- Operate and maintain 6+ analytical instrumentation to ensure proper instrument functionality.
- Research and apply EPA methods to create and update laboratory SOPs.
- Review department analytical data for final client reports ensuring compliance with EPA and other accrediting agency requirements.
- Set up new analytical instrumentation to further enhance laboratory testing methods.
- Research and test scientific methods to improve current SOPs.
- Maintain a working knowledge of all testing procedures in the Inorganic Department to assist laboratory personnel complete daily tasks.

Chemist

Genysis Nutrition Labs, Salt Lake City, UT, April 2014 – July 2016

- Operate and maintain Perkin Elmer NexION 300X ICP-MS instrumentation to provide clients with timely and accurate analytical results.
- Prepare and analyze client samples daily by ICP-MS digestion methods to ensure accuracy in client results.
- Train 6+ laboratory technicians in analytical methods and instrumentation.
- Peer-review analytical data to ensure data is reported in compliance with SOPs.
- Generate laboratory reports and enter client data in laboratory reporting software.
- Ensure cleanliness of lab and lab equipment.

Chemical Analyst

American West Analytical Labs, Salt Lake City, UT, April 2012 – April 2014

- Operate and maintain Agilent 7700 series and Perkin Elmer 6100 DRC ICP-MS instrumentation.
- Prepare and analyze client samples daily by ICP-MS in accordance with EPA methods.
- Review and update SOPs to meet EPA requirements.
- Perform yearly and quarterly certifications for laboratory instrumentation EPA testing requirements.
- Generate laboratory reports and enter client data in laboratory reporting software.
- Ensure all glassware is clean and stored correctly.

PROFESSIONAL MEMBERSHIP

Institute of Electrical and Electronics Engineers, 2024 – Present

Spatial Utah Data Science, 2022 – Present; Webmaster

Society for Freshwater Science, 2021 – Present

Association for Computing Machinery, 2021 – Present

COMMUNITY SERVICE

Impact Trainings, Salt Lake City, UT, 2010 – 2025

Staff for adult and youth trainings

Belay at a ropes course

Exchange Club of Naperville, Naperville, IL, 2000 – 2006

Ribfest Steering Committee Member

LANGUAGES

English: Native Language

Dutch: Novice Listener, Novice Speaker, Novice Reading and Writing